

TADOTSU, Japan

WMO: 478900

Lat: 34.2761N

Lon: 133.7525E

Elev: 18

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.9	33.8	17.6	13.4	38.1	19.7	14.8	39.6	20.8	45.0	18.6	44.2	5.1	150	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	92.6	77.5	90.8	77.1	89.0	76.6	79.7	87.8	78.8	86.9	77.9	86.0	7.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	143.2	84.0	76.5	138.2	83.6	75.5	133.6	83.2	43.3	87.8	42.3	86.9	41.4	86.2	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.0	12.1	DB	29.0	96.0	1.5	1.6	28.0	97.2	27.1	98.2	26.3	99.1	25.2	100.3	(4)
(5)			WB	25.9	80.3	2.0	1.4	24.4	81.2	23.3	82.0	22.2	82.8	20.7	83.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	42.7	43.3	48.8	57.8	66.8	73.3	81.2	83.8	76.8	66.6	55.9	47.0	(6)	
	DBStd	14.83	3.62	4.22	4.88	5.19	3.92	3.72	4.00	3.20	4.84	4.94	5.18	4.37	(7)	
	HDD50	629	227	193	84	5	0	0	0	0	0	0	8	112	(8)	
	HDD65	2915	691	607	503	219	27	0	0	0	0	38	273	557	(9)	
	CDD50	5050	1	5	45	241	522	698	967	1048	804	513	186	20	(10)	
	CDD65	1865	0	0	0	4	84	248	502	583	355	87	2	0	(11)	
	CDH74	16506	0	0	0	18	248	1112	5021	6941	2875	291	0	0	(12)	
	CDH80	5847	0	0	0	1	19	189	1840	2968	804	26	0	0	(13)	
(14) Wind	WSAvg	5.3	6.6	5.9	5.7	5.3	4.8	4.5	4.7	5.0	4.9	5.0	5.3	6.2	(14)	
(15) Precipitation	PrecAvg	44.7	1.6	1.9	3.3	3.2	4.6	6.1	6.2	3.8	5.7	4.1	2.5	1.6	(15)	
	PrecMax	64.2	4.4	3.8	7.9	7.2	10.1	12.0	14.2	11.7	17.7	12.7	6.8	3.7	(16)	
	PrecMin	30.7	0.4	0.5	1.6	1.2	1.6	1.0	1.0	0.6	1.6	1.1	0.7	0.2	(17)	
	PrecStd	9.7	1.0	0.9	1.5	1.3	1.8	2.8	4.1	3.1	3.6	2.7	1.6	0.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	60.8	67.8	76.6	82.4	88.5	94.1	95.5	91.8	83.0	71.6	62.3	(19)	
		MCWB	48.9	53.0	54.6	60.2	64.5	74.4	77.6	77.8	76.3	73.0	62.2	55.9	(20)	
	2%	DB	53.5	55.7	63.0	72.3	79.1	84.5	91.8	93.4	88.9	79.5	69.2	59.3	(21)	
		MCWB	46.3	48.2	52.7	58.6	63.3	71.5	77.0	77.8	75.6	69.4	61.1	52.6	(22)	
	5%	DB	51.5	53.2	59.9	69.3	76.8	81.9	90.1	91.8	86.7	77.1	67.3	57.2	(23)	
		MCWB	44.8	46.7	50.9	57.1	62.6	70.0	76.7	77.6	74.8	67.2	59.8	50.2	(24)	
	10%	DB	49.7	50.9	57.2	66.5	74.5	79.8	88.1	90.1	84.5	75.0	65.1	55.1	(25)	
		MCWB	43.3	44.5	49.1	56.0	62.2	69.3	76.4	77.2	73.7	65.7	58.1	48.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	55.0	57.6	64.6	69.5	75.6	80.1	81.4	79.5	75.0	65.9	57.9	(27)	
		MCDB	53.9	59.9	62.7	70.6	75.3	83.5	88.5	89.7	86.7	80.8	68.9	60.3	(28)	
	2%	WB	47.9	50.7	54.8	61.9	67.4	74.3	79.0	80.1	77.9	72.1	63.5	54.2	(29)	
		MCDB	51.6	54.1	59.9	67.5	73.0	81.4	87.3	88.3	84.8	77.6	67.2	58.0	(30)	
	5%	WB	46.2	47.9	52.9	60.0	66.0	73.0	78.2	79.3	76.6	69.7	61.2	51.7	(31)	
		MCDB	50.3	51.6	57.9	66.0	72.2	79.2	86.4	87.5	83.7	74.8	65.9	55.7	(32)	
	10%	WB	44.1	45.6	50.7	58.3	64.7	71.6	77.2	78.5	75.5	67.2	59.0	49.2	(33)	
		MCDB	49.0	49.9	56.2	64.4	71.3	76.9	85.5	86.7	82.7	73.2	64.1	54.2	(34)	
(35) Mean Daily Temperature Range	MDBR		11.3	12.4	13.8	14.7	14.3	11.4	11.3	12.2	12.2	13.1	12.9	11.8	(35)	
	5% DB	MCDBR	13.9	15.5	17.5	18.7	17.9	14.4	13.7	13.9	13.6	14.1	14.1	14.6	(36)	
		MCWBR	10.1	10.6	10.7	9.2	8.0	5.6	4.4	4.8	5.8	7.5	9.6	10.5	(37)	
	5% WB	MCDBR	13.1	13.3	15.3	14.3	13.2	11.5	12.7	12.8	12.2	11.5	12.4	13.0	(38)	
		MCWBR	10.9	10.4	11.1	8.7	7.3	5.2	4.5	5.0	5.7	7.4	9.5	11.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.396	0.441	0.499	0.528	0.530	0.550	0.551	0.524	0.484	0.442	0.415	0.390	(40)	
	taud		2.173	2.068	1.926	1.907	1.972	1.999	2.049	2.116	2.199	2.230	2.209	2.208	(41)	
	Ebn at Noon		249	250	245	245	246	240	239	244	248	248	241	243	(42)	
	Edh at Noon		37	46	57	61	58	56	53	49	43	39	36	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		779	969	1279	1543	1676	1487	1611	1653	1297	1077	816	686	(44)	
	RadStd		54	75	95	142	160	145	206	164	128	107	69	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+1.21	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.49	N/A	N/A	N/A	N/A

Nomenclature: See separate page